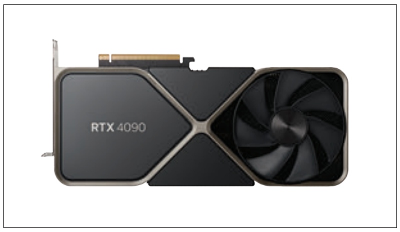
NVIDIA VS AMD

PC GAMING SUPREMACY CHAMPIONS!

Mithun Mohandas | mithun@digit.in

While discrete graphics cards became commonplace a little later, the battle between the two incumbents has played out across internet forums for more than 20 years now. Between AMD and NVIDIA, it was ATI who used to make mainstream graphics cards for PC back in the mid-80s. If that statement sounded confusing that's because the AMD graphics cards that we know today were actually made by ATI before it got acquired by AMD in 2006. NVIDIA came along nearly a decade later in 1993. Each manufacturer would put out a new generation of GPUs every few years and the debates would fire up again. But it wasn't AMD vs NVIDIA back then ... rather they both had to go up against the market leader - 3dfx Voodoo.

The release cycles got narrower only in the mid-90s and since then we've seen a new generation of graphics cards come out every alternate year. Either it was AMD who was nearly as good as 3dfx or sometimes NVIDIA would have the better product. One nasty acquisition by 3dfx sent it barrelling down to bankruptcy and was soon acquired by NVIDIA. The PC ecosystem was dominated by the Apple Macintosh and IBM PCs back then and GPU manufacturers would design for the two camps. ATI's cards were designed as per what Apple wanted whereas NVIDIA was more PC-centric. As the years went by, NVIDIA and AMD emerged as the only



two serious competitors in the consumer graphics card segment. We're not trying to stoke any new flames here but here's the current state of the battle.

HARDWARE

NVIDIA has the upper hand here with their RTX 4090 ruling the roost. It's a ridiculously expensive graphics card and churns out more FPS than the human eye can see. AMD, on the other hand, is a step behind with their Radeon RX 7900 XTX. Not that the AMD card is bad or anything, it's an excellent graphics card for 4K gaming. It's just that NVIDIA has the upper hand with hardware-accelerated ray tracing performance and also energy efficiency. When tested across a variety of video games, the RTX 4090 consumes around 410 Watts to give about 144 FPS whereas the Radeon RX 7900 XTX gives about 116 FPS while consuming 360 Watts. That works out to be about 2.85 Watts per frame for the RTX 4090 and 3.10 Watts per frame for the RX 7900 XTX. It's close but NVIDIA being better at ray-tracing does have the advantage.